

Section 1.3 - Question 3

a

p	q	$p \vee q$	$q \vee p$
True	True	True	True
True	False	True	True
False	True	True	True
False	False	False	False

b

p	q	$p \wedge q$	$q \wedge p$
True	True	True	True
True	False	False	False
False	True	False	False
False	False	False	False

Section 1.3 - Question 7d

Ibrahim is smart and hard working.

Negation: "Ibrahim is not smart or not hard working."

Section 1.3 - Question 9b

Conditional-Disjunction: $p \rightarrow q \equiv \neg p \vee q$

Equivalent Expressions

$(p \rightarrow q) \rightarrow r$
 $\neg(p \rightarrow q) \vee r$
 $\neg(\neg p \vee q) \vee r$
 $(p \wedge \neg q) \vee r$

Section 1.3 - Question 12d

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)$	$((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

Section 1.3 - Question 15b

Expression	Transformation/Reason
$p \rightarrow (p \vee q)$	
$\neg p \vee (p \vee q)$	Conditional-Disjunction Equivalence
$(\neg p \vee p) \vee q$	Associative Law for Disjunction
$\text{True} \vee q$	Negation Law
True	Domination Law

Section 1.3 - Question 15e

Expression	Transformation/Reason
$\neg(p \rightarrow q) \rightarrow p$	
$\neg(\neg p \vee q) \rightarrow p$	Conditional-Disjunction Equivalence
$p \wedge \neg q \rightarrow p$	2nd De Morgan's Law and Double Negation Law
$\neg(p \wedge \neg q) \vee p$	Conditional-Disjunction Equivalence
$\neg p \vee q \vee p$	1st De Morgan's Law
$\neg p \vee p \vee q$	Commutative Law for Disjunction
$\text{True} \vee q$	Negation Law
True	Domination Law

Section 1.3 - Question 17

a

p	q	$p \wedge q$	$p \vee (p \wedge q)$
True	True	True	True
True	False	False	True
False	True	False	False
False	False	False	False

$$p \equiv p \vee (p \wedge q)$$

b

p	q	$p \vee q$	$p \wedge (p \vee q)$
True	True	True	True
True	False	True	True
False	True	True	False
False	False	False	False

$$p \equiv p \wedge (p \vee q)$$

Section 1.3 - Question 19

Expression	Transformation/Reason
$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	
$(\neg q \wedge (\neg p \vee q)) \rightarrow \neg p$	Conditional-Disjunction Equivalence
$(\neg q \wedge \neg p) \vee (\neg q \wedge q)$	Distributive Law
$\rightarrow \neg p$	
$(\neg q \wedge \neg p) \vee \text{False} \rightarrow$	Negation Law
$\neg p$	
$\neg q \wedge \neg p \rightarrow \neg p$	Domination Law
$\neg(\neg q \wedge \neg p) \vee \neg p$	Conditional-Disjunction Equivalence
$q \vee p \vee \neg p$	De Morgan's Law and Double Negation Law
$q \vee \text{True}$	Negation Law
True	Domination Law

$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology.

Section 1.3 - Question 65a

Expression	Transformation/Reason
$(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$	
$(p \vee \neg q) \wedge ((\neg p \vee q) \wedge (\neg p \vee \neg q))$	Associative Law for Conjunction
$(p \vee \neg q) \wedge (\neg p \vee (q \wedge \neg q))$	Distributive Law
$(p \vee \neg q) \wedge (\neg p \vee \text{False})$	Negation Law
$(p \vee \neg q) \wedge \neg p$	Domination Law
$(p \wedge \neg p) \vee (\neg q \wedge \neg p)$	Distributive Law
$\text{False} \vee (\neg q \wedge \neg p)$	Negation Law
$\neg q \wedge \neg p$	Domination Law

Solution: {p: False, q: False}

Section 1.3 - Question 66b

Expression: $(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg s) \wedge (\neg p \vee \neg r \vee \neg s) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg r \vee \neg s)$

The expression is a conjunction of six 3-variable disjunctions. There is at least one negated variable in each of the six disjunctions. Therefore, when all variables are `False`, all six disjunctions will be `True`, and their conjunction (the expression) will be `True`. So, the expression has at least one solution and is therefore satisfiable.

Solution: {p: `False`, q: `False`, r: `False`, s: `False`}

Section 1.4 - Question 1

- a) True
- b) True
- c) False

Section 1.4 - Question 2

- a) True
- b) False
- c) False
- d) True

Section 1.4 - Question 5b

Every student spends more than five hours every weekday in class

Section 1.4 - Question 7

- a) For every person, if they are a comedian, then they are funny.
- b) Every person is both a comedian and funny.
- c) There exists a person that is not funny if they are a comedian.
- d) There exists a person that is both a comedian and funny.

Section 1.4 - Question 8b

Translate these statements into English, where $R(x)$ is “ x is a rabbit” and $H(x)$ is “ x hops” and the domain consists of all animals.

$\forall x (R(x) \text{ and } H(x))$ = Every animal is both a rabbit and hops.

Section 1.4 - Question 9

- a) $\exists x(P(x) \wedge Q(x))$
- b) $\exists x(P(x) \wedge \neg Q(x))$
- c) $\forall x(P(x) \vee Q(x))$
- d) $\neg \exists x(P(x) \vee Q(x))$

Section 1.4 - Question 13

- a) True ($n + 1 > n \equiv 1 > 0$)
- b) True (witness: $n = 0$)
- c) True (witness: $n = 0$)
- d) False (counter-example: $n = -1$)

Section 1.4 - Question 17c

$$\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$$

Section 1.4 - Question 25b

$\neg \forall x P(x)$ where $P(x)$ is "x is perfect" and the domain is everyone.

Section 1.4 - Question 33a

Statement: $\exists x P(x)$ where $P(x)$ is “x can learn new tricks” and the domain is old dogs.

Negation: $\forall x \neg P(x)$ where $P(x)$ is “x can learn new tricks” and the domain is old dogs.

Section 1.4 - Question 36c

Statement: $\exists x(-4 \leq x \leq 1)$

Negation: $\neg \exists x((-4 \leq x) \wedge (x \leq 1)) \equiv \forall x \neg((-4 \leq x) \wedge (x \leq 1)) \equiv$
 $\forall x((-4 > x) \vee (x > 1))$

Without Negation Symbol: $\forall x((-4 > x) \text{ or } (x > 1))$

Section 1.4 - Question 37

- a) No counterexample
- b) counterexample: $x = 0$
- c) counterexample: $x = 0$

Section 1.4 - Question 38b

Statement: $\forall x(x^2 \neq 2)$ where the domain is all real numbers.

Counterexample: $x = \sqrt{2}$

Section 1.4 - Question 43a

At least one mail message, among the nonempty set of messages, can be saved if there is a disk with more than 10 kilobytes of free space

$\exists x((x \in M) \wedge (\exists y F(y) \rightarrow S(x)))$ where M is the set of mail messages, y 's domain is the set of disks, $S(x)$ is "x can be saved", and $F(y)$ is "y has more than 10 kilobytes of free space".

Section 1.4 - Question 53

1. Let $P(x)$ be “ x is even” and $Q(x)$ be “ x is odd” and the domain of x be all numbers.
2. $\exists x P(x) \wedge \exists x Q(x)$ is true because there exists an even number and there exists an odd number.
3. $\exists x (P(x) \wedge Q(x))$ is false because there is no number that is both even and odd.
4. Therefore, $\exists x P(x) \wedge \exists x Q(x)$ is not equivalent to $\exists x (P(x) \wedge Q(x))$.

Section 1.4 - Question 55

- a) $\exists!xP(x) \rightarrow \exists xP(x) \equiv \text{True}$
- b) $\forall xP(x) \rightarrow \exists!xP(x) \equiv \text{False}$ when the domain's size is ≤ 1
- c) $\exists!x\neg P(x) \rightarrow \neg\forall xP(x) \equiv \text{True}$

Section 1.4 - Question 62c

- $P(x)$: x is a clear explanation
- $Q(x)$: x is satisfactory
- $R(x)$: x is an excuse
- $x \in$ the set of all English text

“Some excuses are not clear explanations”

Statement: $\exists x(R(x) \wedge \neg P(x))$

Section 1.4 - Question 63c

- $P(x)$: x is a baby
- $Q(x)$: x is logical
- $R(x)$: x is able to manage a crocodile
- $S(x)$: x is despised
- $x \in$ the set of all people

“Illogical persons are despised”

Statement: $\forall x(\neg Q(x) \rightarrow S(x))$

Section 1.5 - Question 1

- a) For every real number x , there exists a real number y such that x is less than y .
- b) For every real numbers x and y , if both x and y are greater than or equal to zero, then their product xy is greater than or equal to zero.
- c) For all real numbers x and y , there exists a real number z such that the product of x and y is equal to z .

Section 1.5 - Question 4a

- $P(x, y)$: student x has taken class y
- $x \in$ the set of all students in your class
- $y \in$ the set of all computer science classes at your school

Quantification: $\exists x \exists y P(x, y)$

There exists a student in your class who has taken a computer science class at your school.

Section 1.5 - Question 6b

- $C(x, y)$: student x is enrolled in class y
- $x \in$ the set of all students in your school
- $y \in$ the set of all classes being given at your school

Quantification: $\exists x C(x, \text{Math 695})$

There exists a student in your school who is enrolled in the class Math 695.

Section 1.5 - Question 11

- $S(x)$: x is a student
- $F(x)$: x is a faculty member
- $A(x, y)$ x has asked y a question
- $x, y \in$ the set of all people associated with your school

- a) Lois has asked Professor Michaels a question: $A(\text{Lois}, \text{Michaels})$
- b) Every student has asked Professor Gross a question: $\forall x(S(x) \rightarrow A(x, \text{Gross}))$
- c) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller: $\forall x(F(x) \rightarrow (A(x, \text{Miller}) \vee A(\text{Miller}, x)))$
- d) Some student has not asked any faculty member a question: $\exists x(S(x) \wedge \forall y(F(y) \rightarrow \neg A(x, y)))$
- e) There is a faculty member who has never been asked a question by a student: $\exists x(F(x) \wedge \forall y(S(y) \rightarrow \neg A(y, x)))$
- f) Some student has asked every faculty member a question: $\exists x(S(x) \wedge \forall y(F(y) \rightarrow A(x, y)))$
- g) There is a faculty member who has asked every other faculty member a question: $\exists x(F(x) \wedge \forall y(F(y) \rightarrow A(x, y)))$
- h) Some student has never been asked a question by a faculty member: $\exists x(S(x) \wedge \forall y(F(y) \rightarrow \neg A(y, x)))$

Section 1.5 - Question 21

Every positive integer is the sum of the squares of four integers.

- $P(x)$: x is a positive integer
- $Q(x)$: x is the sum of the squares of four integers

Quantification: $\forall x(P(x) \rightarrow Q(x))$ where $x \in$ the set of all integers

Section 1.5 - Question 23

- a) $\forall x \forall y ((P(x) \wedge P(y)) \rightarrow R(x, y))$ where $x, y \in$ the set of all real numbers and $P(x)$ is “ x is a negative” and $R(x, y)$ is “ xy is positive”
- b) $\forall x (x - x = 0)$ where $x \in$ the set of all real numbers
- c) $\forall x (P(x) \rightarrow Q(x))$ where $x \in$ the set of all real numbers and $P(x)$ is “ x is positive” and $Q(x)$ is “ x has exactly two square roots”
- d) $\forall x (P(x) \rightarrow Q(x))$ where $x \in$ the set of all real numbers and $P(x)$ is “ x is a negative” and $Q(x)$ is “ x does not have a square root that is a real number”

Section 1.5 - Question 24d

Quantification: $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

For every real number x and for every real number y , the product of x and y is non-zero if and only if both x and y are non-zero.